

Lab Exercises -2

(React Programs on Event Handling, Props and State and Bootstrap)

1. State Management and Event Handling (10 marks)

Create a React component called Counter that implements a simple counter with the following

features:

i. Display the current count

ii. Include buttons to increment and decrement the count

iii. Add a reset button to set the count back to zero

iv. Implement an input field where users can set a custom count value

Ensure proper state management and event handling. Use Bootstrap for styling the component

Code:

[React Code \(Counter.js\)](#)

Create a new React component for the counter.

```
javascript
CopyEdit
import React, { useState } from "react";
import "bootstrap/dist/css/bootstrap.min.css";

const Counter = () => {
  const [count, setCount] = useState(0);
  const [customValue, setCustomValue] = useState("");

  const increment = () => setCount(count + 1);
  const decrement = () => setCount(count - 1);
  const reset = () => setCount(0);
  const setCustomCount = () => {
    const value = parseInt(customValue, 10);
    if (!isNaN(value)) setCount(value);
  };
};
```

```

return (
  <div className="container mt-5 text-center">
    <h2 className="mb-4">Counter App</h2>
    <h3 className="display-4">{count}</h3>
    <div className="mt-3">
      <button className="btn btn-primary mx-2" onClick={increment}>
        Increment
      </button>
      <button className="btn btn-danger mx-2" onClick={decrement}>
        Decrement
      </button>
      <button className="btn btn-warning mx-2" onClick={reset}>
        Reset
      </button>
    </div>
    <div className="mt-4">
      <input
        type="number"
        className="form-control d-inline w-25"
        value={customValue}
        onChange={ (e) => setCustomValue(e.target.value) }
        placeholder="Set Custom Count"
      />
      <button className="btn btn-success mx-2" onClick={setCustomCount}>
        Set Count
      </button>
    </div>
  </div>
);
};

export default Counter;

```

2. Parent-Child Communication (10 marks)

Create a parent component called **TodoList** and a child component called **TodoItem**. Implement the

following:

- i. **TodoList** should maintain a list of todos in its state
- ii. **TodoItem** should display a todo item and have a button to mark it as complete
- iii. When a todo is marked as complete, communicate this change back to the parent
- iv. Use **Bootstrap** to style the list and items

Code:

1 **TodoList.js** (Parent Component)

Manages the list of todos and updates state when an item is marked as complete.

```
javascript
CopyEdit
import React, { useState } from "react";
import "bootstrap/dist/css/bootstrap.min.css";
import TodoItem from "../TodoItem";

const TodoList = () => {
  const [todos, setTodos] = useState([
    { id: 1, text: "Complete React assignment", completed: false },
    { id: 2, text: "Review React props and state", completed: false },
    { id: 3, text: "Learn Bootstrap styling", completed: false },
  ]);

  const markComplete = (id) => {
    setTodos(
      todos.map((todo) =>
        todo.id === id ? { ...todo, completed: true } : todo
      )
    );
  };

  return (
    <div className="container mt-5">
```

```

    <h2 className="mb-4 text-center">Todo List</h2>
    <ul className="list-group">
      {todos.map((todo) => (
        <TodoItem key={todo.id} todo={todo}
markComplete={markComplete} />
      ))}
    </ul>
  </div>
);
};

export default TodoList;

```

2 TodoItem.js (Child Component)

Displays a single todo item and marks it as complete when clicked.

```

javascript
CopyEdit
import React from "react";

const TodoItem = ({ todo, markComplete }) => {
  return (
    <li className={`list-group-item d-flex justify-content-between
align-items-center
  ${todo.completed ? "list-group-item-success" : ""}`>
      <span>{todo.text}</span>
      <button
        className="btn btn-sm btn-success"
        onClick={() => markComplete(todo.id)}
        disabled={todo.completed}
      >
        {todo.completed ? "Completed" : "Mark Complete"}
      </button>
    </li>
  );
};

export default TodoItem;

```

3. Stateless Components and Props vs State (10 marks)

Create a React application that demonstrates the use of both stateful and stateless components:

- i. Implement a stateful **UserProfile** component that manages user information (name, email, bio)
- ii. Create a stateless **DisplayProfile** component that receives user information as props and displays it
- iii. Add an edit button in **UserProfile** that toggles between display and edit modes
- iv. Use **Bootstrap** for styling, including a form for editing and a card for displaying the profile

Here's the React code for **UserProfile** (stateful) and **DisplayProfile** (stateless) components, implementing **props vs. state**, **edit mode toggle**, and **Bootstrap styling**.

1 UserProfile.js (Stateful Component)

Manages the user information and toggles between display and edit modes.

```
javascript
CopyEdit
import React, { useState } from "react";
import "bootstrap/dist/css/bootstrap.min.css";
import DisplayProfile from "../DisplayProfile";

const UserProfile = () => {
  const [user, setUser] = useState({
    name: "John Doe",
    email: "johndoe@example.com",
    bio: "Software Developer | React Enthusiast",
  });

  const [isEditing, setIsEditing] = useState(false);
  const [formData, setFormData] = useState(user);

  const handleChange = (e) => {
```

```

    setFormData({ ...formData, [e.target.name]: e.target.value });
  };

const handleSave = () => {
  setUser(formData);
  setIsEditing(false);
};

return (
  <div className="container mt-5">
    <h2 className="text-center mb-4">User Profile</h2>
    <div className="card p-4">
      {isEditing ? (
        <div>
          <div className="mb-3">
            <label className="form-label">Name</label>
            <input
              type="text"
              className="form-control"
              name="name"
              value={formData.name}
              onChange={handleChange}
            />
          </div>
          <div className="mb-3">
            <label className="form-label">Email</label>
            <input
              type="email"
              className="form-control"
              name="email"
              value={formData.email}
              onChange={handleChange}
            />
          </div>
          <div className="mb-3">
            <label className="form-label">Bio</label>
            <textarea
              className="form-control"
              name="bio"
              value={formData.bio}
              onChange={handleChange}
            />
          </div>
          <button className="btn btn-success" onClick={handleSave}>
            Save
          </button>
        </div>
      ) : (
        <DisplayProfile user={user} />
      )}
      <button
        className="btn btn-primary mt-3"
        onClick={() => setIsEditing(!isEditing)}
      >
        {isEditing ? "Cancel" : "Edit Profile"}
      </button>
    </div>
  </div>

```

```
        </div>
    );
};

export default UserProfile;
```

2 DisplayProfile.js (Stateless Component)

Displays the user information passed as props.

```
javascript
CopyEdit
import React from "react";

const DisplayProfile = ({ user }) => {
    return (
        <div>
            <h4>{user.name}</h4>
            <p>Email: {user.email}</p>
            <p>Bio: {user.bio}</p>
        </div>
    );
};

export default DisplayProfile;
```

4. Async State Initialization and Bootstrap Integration (10 marks)

Create a React component called ProductList that fetches product data asynchronously and displays it

in a Bootstrap table:

- i. Use useState and useEffect hooks for state management and async initialization**
- ii. Fetch product data from a mock API (you can use setTimeout to simulate an API call)**
- iii. Display the products in a Bootstrap table with columns for ID, Name, Price, and Category**
- iv. Implement a Bootstrap navbar with a search input that filters the products based on name**

ProductList.js (Main Component)

Fetches product data asynchronously and displays it in a Bootstrap table.

```
javascript
CopyEdit
import React, { useState, useEffect } from "react";
import "bootstrap/dist/css/bootstrap.min.css";

const ProductList = () => {
  const [products, setProducts] = useState([]);
  const [searchQuery, setSearchQuery] = useState("");

  // Simulating an API call using setTimeout
  useEffect(() => {
    setTimeout(() => {
      const mockProducts = [
        { id: 1, name: "Laptop", price: 70000, category: "Electronics" },
        { id: 2, name: "Smartphone", price: 30000, category: "Electronics" },
        { id: 3, name: "Table", price: 8000, category: "Furniture" },
        { id: 4, name: "Headphones", price: 5000, category: "Accessories" },
        { id: 5, name: "Shoes", price: 2000, category: "Fashion" },
      ];
      setProducts(mockProducts);
    }, 2000); // Simulating network delay
  }, []);

  // Filter products based on search query
  const filteredProducts = products.filter((product) =>
    product.name.toLowerCase().includes(searchQuery.toLowerCase())
  );

  return (
    <div className="container mt-4">
      {/* Bootstrap Navbar */}
      <nav className="navbar navbar-light bg-light mb-3">
        <div className="container-fluid">
          <a className="navbar-brand">Product List</a>
          <input
            type="text"
            className="form-control w-25"
            placeholder="Search by name..."
            value={searchQuery}
            onChange={(e) => setSearchQuery(e.target.value)}
          />
        </div>
      </nav>

      {/* Bootstrap Table */}
      <div className="table-responsive">
        <table className="table table-bordered table-striped">
          <thead className="table-dark">
            <tr>
              <th>ID</th>
              <th>Name</th>
              <th>Price (₹)</th>
            </tr>
          </thead>
          <tbody>
            {filteredProducts.map((product) => (
              <tr>
                <td>{product.id}</td>
                <td>{product.name}</td>
                <td>{product.price}</td>
              </tr>
            ))}
          </tbody>
        </table>
      </div>
    </div>
  );
}
```

```

        <th>Category</th>
      </tr>
    </thead>
    <tbody>
      {filteredProducts.length > 0 ? (
        filteredProducts.map((product) => (
          <tr key={product.id}>
            <td>{product.id}</td>
            <td>{product.name}</td>
            <td>{product.price}</td>
            <td>{product.category}</td>
          </tr>
        ))
      ) : (
        <tr>
          <td colspan="4" className="text-center">
            No products found
          </td>
        </tr>
      )}
    </tbody>
  </table>
</div>
</div>
);
};

export default ProductList;

```

5. Component Hierarchy and Bootstrap Modals (10 marks)

Design a React application for a simple e-commerce product page with the following component

hierarchy:

- **ProductPage (main component)**
- **ProductImage**
- **ProductInfo**
- **AddToCartButton**

Use Bootstrap for styling all components and implementing the model.

Implement the following features:

- i. **ProductPage** should manage the overall state of the product
- ii. **ProductImage** should display the product image

- iii. **ProductInfo** should show product details (name, price, description)
 - iv. **AddToCartButton** should trigger a **Bootstrap** modal when clicked
 - v. The modal should display a summary of the product and have options to proceed to checkout
- or continue shopping

Here's the React implementation for **ProductPage**, demonstrating **component hierarchy**, **state management**, **Bootstrap styling**, and **a modal for the Add to Cart button**.

Code

1 ProductPage.js (Main Component)

Manages product state and handles the Bootstrap modal.

```
javascript
CopyEdit
import React, { useState } from "react";
import "bootstrap/dist/css/bootstrap.min.css";
import ProductImage from "../ProductImage";
import ProductInfo from "../ProductInfo";
import AddToCartButton from "../AddToCartButton";

const ProductPage = () => {
  const product = {
    id: 1,
    name: "Wireless Headphones",
    price: 4999,
    description: "High-quality wireless headphones with noise cancellation.",
    image: "https://via.placeholder.com/300", // Placeholder image URL
  };

  const [showModal, setShowModal] = useState(false);

  return (
    <div className="container mt-5">
      <h2 className="text-center mb-4">Product Page</h2>
      <div className="row">
        <div className="col-md-6">
          <ProductImage image={product.image} name={product.name} />
        </div>
        <div className="col-md-6">
          <ProductInfo product={product} />
          <AddToCartButton setShowModal={setShowModal} />
        </div>
      </div>
    </div>
  );
};
```

```

    </div>

    { /* Bootstrap Modal */ }
    { showModal && (
      <div className="modal fade show d-block" tabIndex="-1">
        <div className="modal-dialog">
          <div className="modal-content">
            <div className="modal-header">
              <h5 className="modal-title">Added to Cart</h5>
              <button
                type="button"
                className="btn-close"
                onClick={() => setShowModal(false)}
              ></button>
            </div>
            <div className="modal-body">
              <p>
                <strong>{product.name}</strong> has been added to your
cart.

                </p>
                <p>Price: ₹{product.price}</p>
              </div>
              <div className="modal-footer">
                <button
                  className="btn btn-secondary"
                  onClick={() => setShowModal(false)}
                >
                  Continue Shopping
                </button>
                <button className="btn btn-primary">Proceed to
Checkout</button>
              </div>
            </div>
          </div>
        </div>
      </div>
    ) }
  </div>
);
};

export default ProductPage;

```

2 ProductImage.js (Displays Product Image)

javascript

CopyEdit

```
import React from "react";
```

```
const ProductImage = ({ image, name }) => {
  return <img src={image} alt={name} className="img-fluid rounded shadow" />;
};
```

```
export default ProductImage;
```

3 ProductInfo.js (Displays Product Details)

javascript

CopyEdit

```
import React from "react";

const ProductInfo = ({ product }) => {
  return (
    <div>
      <h3>{product.name}</h3>
      <h4 className="text-success">₹{product.price}</h4>
      <p>{product.description}</p>
    </div>
  );
};

export default ProductInfo;
```

4 AddToCartButton.js (Triggers Modal)

javascript

CopyEdit

```
import React from "react";

const AddToCartButton = ({ setShowModal }) => {
  return (
    <button
      className="btn btn-warning mt-3"
      onClick={() => setShowModal(true)}
    >
      Add to Cart
    </button>
  );
};

export default AddToCartButton;
```